



2017 Preliminary Half Yearly Examination

Mathematics Extension I

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour and 30 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 54

Section I Pages 2 – 4

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 5 – 8

44 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 15 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer page in the writing booklet for Questions 1 – 10.

- 1 How many numbers greater than 40000 can be formed with the digits 2, 3, 4, 5 and 6 if no digit is used more than once?

(A) 48 (B) 72 (C) 96 (D) 120

- 2 In how many ways can a family of eight sit around a circular table if the two youngest family members want to sit together?

(A) $7!$ (B) $2 \times 5!$ (C) $2 \times 6!$ (D) $2 \times 7!$

- 3 What is the domain of the function $y = \frac{1}{\sqrt{4 - x^2}}$?

(A) $x > 2$ (B) $x < 2$
(C) $-2 < x < 2$ (D) $x < -2$ or $x > 2$

- 4 If $2\sec\theta + 3 = 0$, and $\cot\theta > 0$, what is the exact value of $\sin\theta$?

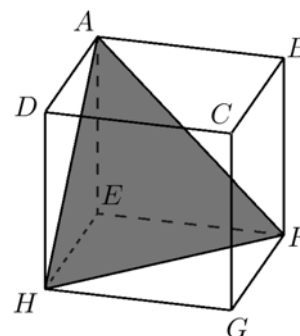
(A) $-\frac{\sqrt{5}}{3}$ (B) $-\frac{\sqrt{5}}{2}$ (C) $\frac{\sqrt{5}}{3}$ (D) $\frac{\sqrt{5}}{2}$

5 How many value(s) of θ , for $0^\circ \leq \theta \leq 360^\circ$, satisfy the equation $\sin \theta \cos \theta = \sin \theta$?

- (A) 2 (B) 3 (C) 4 (D) 5
-

6 $ABCDEFGH$ is a cube of side length p , what is the area of $\triangle AFH$ in terms of p ?

- (A) $\frac{p^2\sqrt{3}}{4}$ (B) $\frac{p^2\sqrt{3}}{2}$
(C) $\frac{p^2}{4}$ (D) $\frac{p^2}{2}$



7 For $0^\circ \leq \alpha \leq 90^\circ$, the least value of $\frac{30}{3\sin^2 \alpha + 2\sin^2 (90^\circ - \alpha)}$ is:

- (A) 15 (B) 10 (C) 6 (D) 5
-

8 When solving $\frac{x-1}{\sqrt{x}} > \frac{2}{x-1}$ within the natural domain, three students obtain the following inequalities:

Student 1: $(x-1)^2 > 2\sqrt{x}$

Student 2: $(x-1)^3 > 2(x-1)\sqrt{x}$

Student 3: $(x-1)^3\sqrt{x} > 2x(x-1)$

Which student(s) will obtain the correct solution to the original inequality?

- (A) Student 1 only (B) Student 2 only
(C) Student 3 only (D) Both students 2 and 3

9 If $f(x+1) = x^4 - 2x + 1$, what is the value of $f(0)$?

(A) 0

(B) 1

(C) 2

(D) 4

10 Given that n and r are both positive integers such that $n \geq r$, which of the following is **NOT** always true?

(A) ${}^nP_1 = {}^nC_1$

(B) ${}^nP_n = {}^nC_n$

(C) ${}^nP_r = {}^nC_r \times r!$

(D) ${}^nP_r \geq {}^nC_r$

Section II

44 marks

Attempt Questions 11 – 14

Allow about 1 hour and 15 minutes for this section

Answer each question in the appropriate section of the writing booklet. Extra writing paper is available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (11 marks) Use the Question 11 section of the writing booklet.

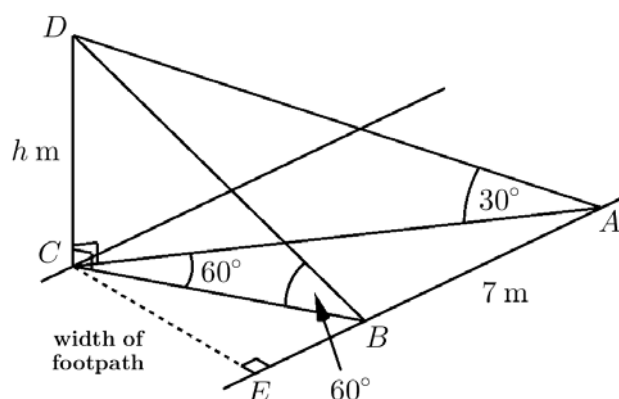
- (a) Solve $\frac{x}{x-4} \leq 5$. **3**
- (b) The letters A, E, I, O and U are vowels. In how many ways can the letters of the word **MATHEMATICS** be arranged in a line if:
- (i) there are no restrictions? **1**
 - (ii) the vowels are all together? **1**
 - (iii) both the letter **M**'s must be immediately followed by the letter **A**? **1**
- (c)
- (i) Solve $x^4 - 4x^2 + 3 = 0$. **1**
 - (ii) Hence, or otherwise, solve $\tan^4 \theta - 4 \tan^2 \theta + 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. **2**
- (d) Sketch the region on the Cartesian plane satisfying the inequality $y < -\sqrt{x-1}$. **2**

End of Question 11

Question 12 (11 marks) Use the Question 12 section of the writing booklet.

- (a) A footpath on horizontal ground has two parallel edges. CD is a vertical flagpole of height h metres which stands with its base, C , on one edge of the footpath. A and B are two points on the other edge of the footpath such that $AB = 7$ m and $\angle ACB = 60^\circ$.

From A and B , the angles of elevation of the top of the flagpole, D , are 30° and 60° respectively.



- | | |
|---|----------|
| (i) Show that $BC = \frac{h}{\sqrt{3}}$. | 1 |
| (ii) Find a similar expression for AC . | 1 |
| (iii) Hence, find the exact height of the flagpole. | 2 |
| (iv) Find the exact width of the footpath. | 3 |
-
- (b) From a group of 6 men and 8 women, a committee of 5 people is to be chosen, how many ways can this committee be formed if:
- | | |
|---|----------|
| (i) there are no restrictions? | 1 |
| (ii) the committee consists of 2 men and 3 women? | 1 |
| (iii) the committee must have at least 1 woman? | 1 |
| (iv) the entire committee is of the same gender? | 1 |

End of Question 12

Question 13 (11 marks) Use the Question 13 section of the writing booklet.

(a) Consider the function $f(x) = x^2 - |2x| - 3$.

(i) Show that $f(x)$ is an even function. 1

(ii) Hence, or otherwise, sketch the graph of $y = f(x)$, showing all intercepts. 2

(iii) What is the range of $y = f(x)$? 1

(b)

(i) On the same set of axes, sketch the graphs of $y = |2x - 4|$ and $y = |x + 1|$, showing all intercepts. 2

(ii) Hence, or otherwise, solve the inequality $|2x - 4| \leq |x + 1|$. 2

(c)

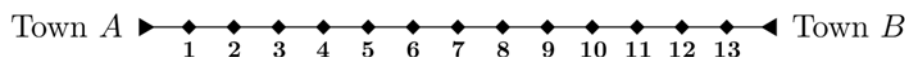
(i) Fully factorise $64k^6 - 1$ as a difference of two squares and as a difference of two cubes. 2

(ii) Hence, or otherwise, factorise $16k^4 + 4k^2 + 1$. 1

End of Question 13

Question 14 (11 marks) Use the Question 14 section of the writing booklet.

- (a) A train is leaving Town A, heading towards Town B without turning around. There are 13 train stations between the two towns:



Key : ♦ Station

- (i) In how many ways can the train stop at 4 of the 13 stations? 1
- (ii) In how many ways can the train stop at 4 of the 13 stations if the train does not stop at consecutive stations? 1

(b) Show that $\frac{(\sin^2 \alpha - \cos^2 \alpha)(1 - \sin \alpha \cos \alpha)}{\cos \alpha (\sec \alpha - \operatorname{cosec} \alpha)(\sin^3 \alpha + \cos^3 \alpha)} = \sin \alpha$. 3

- (c) It can be shown that:

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} \quad (\text{DO NOT PROVE THIS.})$$

- (i) For any $\triangle ABC$, explain why $\tan A + \tan B + \tan C = \tan A \tan B \tan C$. 2
- (ii) It is given, for $\triangle XYZ$, that $\frac{\tan X}{5} = \frac{\tan Y}{6} = \frac{\tan Z}{7} = k$ for some constant k , 3
- show that $k = \sqrt{\frac{3}{35}}$.

- (iii) Hence calculate the size of the smallest angle in $\triangle XYZ$ correct to the nearest minute. 1

End of Paper



YEAR 11 HALF YEARLY EXAMINATION 2017
MATHEMATICS EXTENSION 1
MARKING GUIDELINES

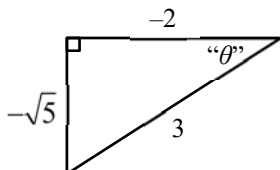
Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	C
4	A
5	B

Question	Answer
6	B
7	B
8	D
9	D
10	B

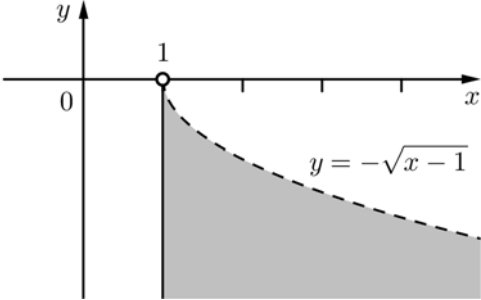
Questions 1 – 10

Sample solution	
1.	Number over 40000 = $3 \times 4!$ $= 72$ $\therefore 72$ numbers are greater than 40000.
2.	2 ways to arrange the two youngest family members $6!$ ways to arrange the remaining six family members as well as the group of two around a circular table.
3.	$4 - x^2 > 0$ $4 > x^2$ $\therefore -2 < x < 2$
4.	$\cot \theta > 0$ if $\tan \theta > 0$ $2 \sec \theta + 3 = 0$ $\sec \theta = \frac{-3}{2}$ $\cos \theta = \frac{2}{-3}$ $\therefore \cos \theta < 0$ and $\tan \theta > 0$ (i.e. θ is in quadrant 3) $\sin \theta = -\frac{\sqrt{5}}{3}$ 
5.	$\sin \theta \cos \theta = \sin \theta$ $\sin \theta (\cos \theta - 1) = 0$ $\therefore \sin \theta = 0$ or $\cos \theta = 1$ For $0^\circ \leq \theta \leq 360^\circ$, $\theta = 0^\circ, 180^\circ, 360^\circ$
6.	$AH = AF = FH$ (equal diagonals of the square faces) ie. $\triangle AFH$ is equilateral. By Pythagoras' theorem, $AH = p\sqrt{2}$. Using the sine rule for area, $A_{\triangle AFH} = \frac{1}{2} \times p\sqrt{2} \times p\sqrt{2} \times \sin 60^\circ$ $= \frac{p^2 \sqrt{3}}{2}$

7.	$\frac{30}{3\sin^2 \alpha + 2\sin^2 (90^\circ - \alpha)} = \frac{30}{3\sin^2 \alpha + 2\cos^2 \alpha}$ $= \frac{30}{\sin^2 \alpha + 2}$ Least value occurs when denominator is largest, i.e. $\sin^2 \alpha = 1$. Least value of $\frac{30}{\sin^2 \alpha + 2} = \frac{30}{1+2}$ $= 10$
8.	Student 1 multiplied both sides by $(x-1)$, which could be negative, the inequality sign may needed to have been flipped. Student 2 multiplied both sides by $\sqrt{x}(x-1)^2$, which is positive, the inequality is ok. Student 3 multiplied both sides by $(\sqrt{x})^2(x-1)^2$, which is positive, the extra $\sqrt{x}$ does not introduce any extra solutions in the natural domain, the inequality is ok.
9.	Let $x = -1$: $f(0) = (-1)^4 - 2 \times (-1) + 1$ $= 1 + 2 + 1$ $= 4$
10.	The number of ways of choosing 1 item from a set of n items does not depend on order, so ${}^n P_1 = {}^n C_1$. By definition, ${}^n C_r = \frac{n!}{r!(n-r)!}$ $= \frac{1}{r!} {}^n P_r$ ${}^n C_r \times r! = {}^n P_r$ The number of ways of choosing r items from a set of n items where order is important is always more than the number of ways of choosing r items from a set of n items where order is not important, i.e. ${}^n P_r \geq {}^n C_r$, with equality when $r = 1$. By elimination, ${}^n P_n \neq {}^n C_n$ is not always true. (Note: For the given conditions, this statement is only true for $n = 1$.)

Section II

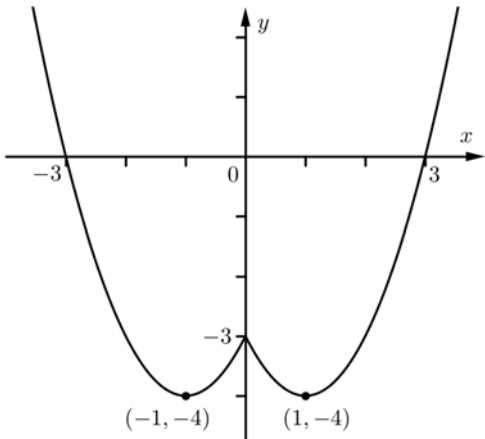
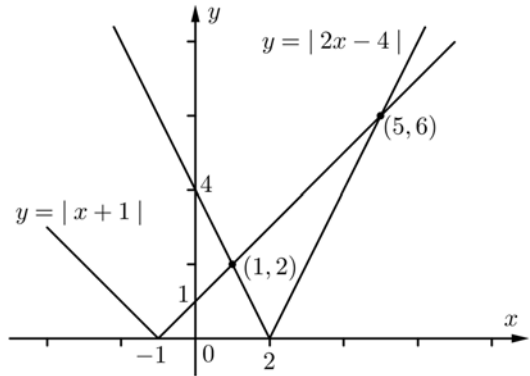
Question 11

Sample solution	Suggested marking criteria
(a) Natural domain for $\frac{x}{x-4} : x \neq 4$ $\frac{x}{x-4} \leq 5$ $x(x-4) \leq 5(x-4)^2$ $0 \leq (x-4)[5(x-4)-x]$ $0 \leq 4(x-4)(x-5)$ $\therefore x < 4 \text{ or } x \geq 5$	• 3 – correct solution • 2 – correctly identifies the two critical points $x = 4$ and $x = 5$ • 1 – attempts to solve the inequation using a suitable method – recognises a restriction in the domain
(b) (i) $\frac{11!}{2!2!2!} = 4\,989\,600$ (ii) There are $\frac{4!}{2!} = 12$ ways of arranging the vowels. Treating the vowels as a single entity, the number of ways to arrange the remaining 7 letters and the single group of vowels is: $\frac{8!}{2!2!} = 10\,080$ Total number of arrangements = $10\,080 \times 12$ = 120 960 (iii) Treating MA as a single entity, the number of ways to arrange (MA)THE(MA)TICS in a line is: $\frac{9!}{2!2!} = 90\,720$	• 1 – correct answer, or equivalent numerical expression • 1 – correct answer, or equivalent numerical expression • 1 – correct answer, or equivalent numerical expression
(c) (i) Let $u = x^2$: $u^2 - 4u + 3 = 0$ $(u-3)(u-1) = 0$ $u = 1 \quad \text{or} \quad u = 3$ $x^2 = 1 \quad \quad x^2 = 3$ $x = \pm 1 \quad \quad x = \pm\sqrt{3}$ (ii) Using the results from (i), and letting $x = \tan \theta$: $\tan \theta = \pm 1 \quad \quad \text{or} \quad \tan \theta = \pm\sqrt{3}$ $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ \quad \quad \theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$ $\therefore \theta = 45^\circ, 60^\circ, 120^\circ, 135^\circ, 225^\circ, 240^\circ, 300^\circ, 315^\circ$	• 1 – correct solution • 2 – correct solution • 1 – correct attempt at solving an appropriate trigonometric equation
(d) 	• 2 – correct region and boundaries • 1 – significant progress towards the correct region and boundaries

Question 12

Sample solution		Suggested marking criteria
(a)	(i) In $\triangle DBC$, $\tan 60^\circ = \frac{h}{BC}$ $BC = \frac{h}{\tan 60^\circ}$ $= \frac{h}{\sqrt{3}}$	1 – correct solution
	(ii) In $\triangle DAC$, $\tan 30^\circ = \frac{h}{AC}$ $AC = \frac{h}{\tan 30^\circ}$ $= h\sqrt{3}$	1 – correct solution
	(iii) In $\triangle ABC$, $AB^2 = BC^2 + AC^2 - 2 \times BC \times AC \times \cos 60^\circ$ $49 = \left(\frac{h}{\sqrt{3}}\right)^2 + (h\sqrt{3})^2 - 2 \times \frac{h}{\sqrt{3}} \times h\sqrt{3} \times \frac{1}{2}$ $49 = \frac{h^2}{3} + 3h^2 - h^2$ $49 = \frac{7h^2}{3}$ $21 = h^2$ $h = \sqrt{21} \text{ (since } h > 0\text{)}$	2 – correct solution 1 – uses cosine rule, showing appropriate substitution – finds AC in terms of h
	(iv) By sine area rule, $A_{\triangle ABC} = \frac{1}{2} \times \frac{h}{\sqrt{3}} \times h\sqrt{3} \times \sin 60^\circ$ $= \frac{h^2}{2} \times \frac{\sqrt{3}}{2}$ $= \frac{21\sqrt{3}}{4}$	3 – correct solution 2 – finds the area of $\triangle ABC$ – significant progress towards solution using $\cos \angle CBA = \frac{1}{2\sqrt{7}}$ or $\sin \angle CAB = \frac{\sqrt{21}}{14}$, or equivalent merit 1 – attempts to use the sine area rule to find the area of $\triangle ABC$
(b)	(i) ${}^{14}C_5 = 2002$	1 – correct answer, or equivalent numerical expression
	(ii) ${}^6C_2 \times {}^8C_3 = 840$	1 – correct answer, or equivalent numerical expression
	(iii) There are ${}^6C_5 = 6$ committees that consist of men only. Therefore, there are $2002 - 6 = 1996$ committees with at least 1 woman.	1 – correct answer, or equivalent numerical expression
	(iv) There are ${}^6C_5 = 6$ committees that consist of men only. There are ${}^8C_5 = 56$ committees that consist of women only. Therefore, there are $56 + 6 = 62$ committees that are entirely made up of members of the same gender.	1 – correct answer, or equivalent numerical expression

Question 13

Sample solution		Suggested marking criteria
(a)	(i) $f(-x) = (-x)^2 - -2x - 3$ $= x^2 - 2x - 3$ (since $ -A = A $ for all A) $= f(x)$ Therefore, $f(x)$ is an even function.	• 1 – correct solution
	(ii) 	• 2 – correct solution • 1 – correctly sketches one branch of $y = f(x)$
	(iii) The vertex of the parabola $y = x^2 - 2x - 3$ is $(1, -4)$. By symmetry of the even function, the range of $f(x) = x^2 - 2x - 3$ is $y \geq -4$.	• 1 – correct solution
(b)	(i) 	• 2 – correct graphs • 1 – one correct graph
	(ii) $1 \leq x \leq 5$	• 2 – correct solution • 1 – correctly identifies two critical points
(c)	(i) By difference of two cubes, $64k^6 - 1 = (4k^2 - 1)(16k^4 + 4k^2 + 1)$ $= (2k + 1)(2k - 1)(16k^4 + 4k^2 + 1)$ By difference of two squares, $64k^6 - 1 = (8k^3 + 1)(8k^3 - 1)$ $= (2k + 1)(4k^2 - 2k + 1)(2k - 1)(4k^2 + 2k + 1)$	• 2 – correct solution • 1 – correct expression for one of the factorisations
	(ii) $16k^4 + 4k^2 + 1 = (4k^2 - 2k + 1)(4k^2 + 2k + 1)$	• 1 – correct solution

Question 14

Sample solution		Suggested marking criteria
(a)	(i) ${}^{13}C_4 = 715$	• 1 – correct answer, or equivalent numerical expression
	(ii) Let S represent a “stop” and P represent a “pass”. The problem is equivalent to arranging 9 P ’s and 4 S ’s so that no S is adjacent to one another. With 9 P ’s, there are 10 spaces that the 4 S ’s could go: $_P_P_P_P_P_P_P_P_P_P_$ Number of ways = ${}^{10}C_4$ $= 210$	• 1 – correct answer, or equivalent numerical expression
(b)	(i) $\begin{aligned} \text{LHS} &= \frac{(\sin^2 \alpha - \cos^2 \alpha)(1 - \sin \alpha \cos \alpha)}{\cos \alpha (\sec \alpha - \operatorname{cosec} \alpha)(\sin^3 \alpha + \cos^3 \alpha)} \\ &= \frac{(\cancel{\sin \alpha + \cos \alpha})(\sin \alpha - \cos \alpha)(1 - \cancel{\sin \alpha \cos \alpha})}{\cos \alpha (\sec \alpha - \operatorname{cosec} \alpha)(\cancel{\sin \alpha + \cos \alpha})(\sin^2 \alpha - \cancel{\sin \alpha \cos \alpha} + \cos^2 \alpha)} \\ &= \frac{\sin \alpha - \cos \alpha}{\cos \alpha \left(\frac{1}{\cos \alpha} - \frac{1}{\sin \alpha} \right)} \\ &= \frac{\cancel{\sin \alpha - \cos \alpha}}{\cancel{\cos \alpha} \left(\frac{\cancel{\sin \alpha - \cos \alpha}}{\sin \alpha \cancel{\cos \alpha}} \right)} \\ &= \sin \alpha \\ &= \text{RHS} \end{aligned}$	• 3 – correct solution • 2 – significant progress towards a valid solution • 1 – correctly factorises the difference of two squares – correctly factorises the sum of two cubes – correctly simplifies $\cos \alpha (\sec \alpha - \operatorname{cosec} \alpha)$
(c)	(i) In any $\triangle ABC$, $A + B + C = 180^\circ$ $\therefore \tan(A + B + C) = 0$ $\frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = 0$ $\tan A + \tan B + \tan C - \tan A \tan B \tan C = 0$ $\tan A + \tan B + \tan C = \tan A \tan B \tan C$	• 2 – correct solutions • 1 – recognising $\tan(A + B + C) = 0$
	(ii) $\frac{\tan X}{5} = \frac{\tan Y}{6} = \frac{\tan Z}{7} = k$ Therefore, $\tan X = 5k$, $\tan Y = 6k$ and $\tan Z = 7k$. $\tan X + \tan Y + \tan Z = \tan X \tan Y \tan Z$ $5k + 6k + 7k = 5k \times 6k \times 7k$ $18k = 210k^3$ $0 = 6k(35k^2 - 3)$ $k \neq 0$ as this implies $\tan X = \tan Y = \tan Z = 0$, which yields a degenerate triangle. $k \neq 0$ as X, Y and Z cannot be all obtuse, therefore $k = \sqrt{\frac{3}{35}}$	• 3 – correct solution • 2 – justifies why k is positive • 1 – attempts to solve for k using an appropriate method
	(iii) $\tan X = 5\sqrt{\frac{3}{35}}$ $X = 55^\circ 40'$	• 1 – correct solution